作业1：（第5次上课时交）

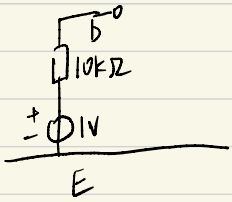
P121 1.3.1 1.3.4 1.3.7 1.3.8 1.3.10 1.3.12 1.3.14 1.3.15

P203 2.1.9（2级） 2.1.13

P148 1.4.5 1.4.7

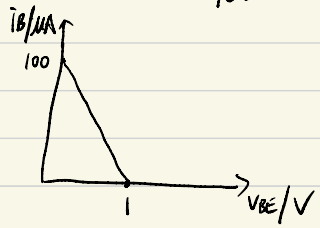
1.3.1 b极左等效为 $U_{eq}=\frac{12}{(110+10)}\cdot 10=1V$

$R_{eq}=110k\Omega\,//\,10k\Omega\approx 10k\Omega$



$V_{BE}=1-\frac{i_B(\mu A)}{100}$ $i_B=100-100V_{BE}$

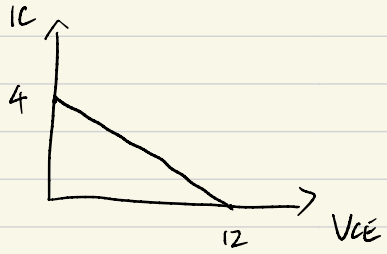
$\therefore I_{BQ}\approx 30\mu A$ $V_{BEQ}\approx 0.7V$



$V_{CE}=V_{CC}-I_CR_C$

$\therefore V_{CE}=12-3i_C(mA)$

$i_C=4-\frac{1}{3}V_{CE}$



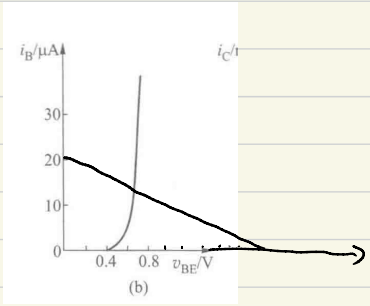
$I_{CQ}\approx 2.2mA$

$V_{CE}\approx 6V$

1.3.4 (1) $V_{BE}=V_{BB}-i_BR_b$

$\therefore V_{BE}=2-\frac{i_B(\mu A)}{10}$ $i_B=20-10V_{BE}$

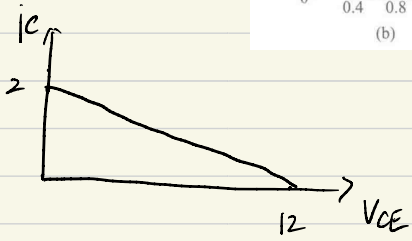
$\therefore I_{BQ}\approx 14\mu A$



$V_{CC}=V_{CE}+R_ci_C$

$\therefore V_{CE}=12-6i_C$

$\therefore I_{CQ}\approx 0.69mA$ $V_{CEQ}\approx 6.9V$



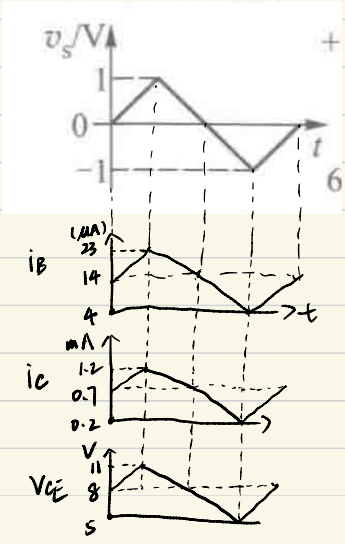
(2) $V_S=1V$, $V_{CE}=3-\frac{i_B}{10}$

$V_S=-1V$, $V_{CE}=|-\frac{i_B}{10}$

(3) 最大幅度 $\Delta V_O=11-5\ V=6V$

$V_{CEQ}=6V$ $I_{BQ}\approx 20\mu A$

$R_b=65k\Omega$

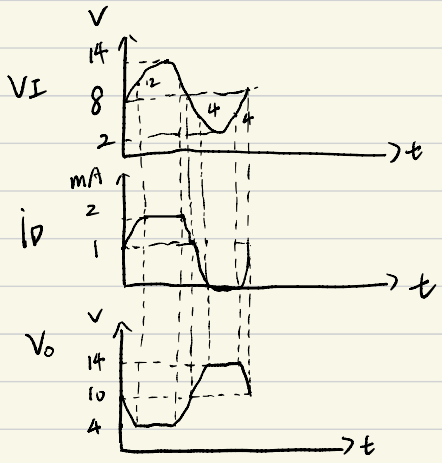


1.3.7 (1) $V_{DS} = 15 - 5i_D$

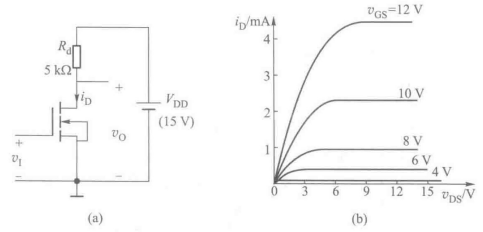
$V_I = 2V$ 截止区　$V_I = 6V$，放大区　$V_I = 10V$ 可变电阻区

$V_I = 4V$ 截止区　$V_I = 8V$，放大区　$V_I = 12V$ 可变电阻区

(2) $V_I = 8 + 6\sin \omega t$

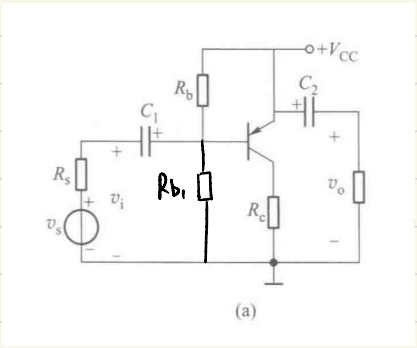


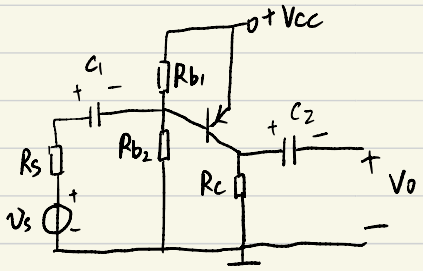
图题 1.3.6



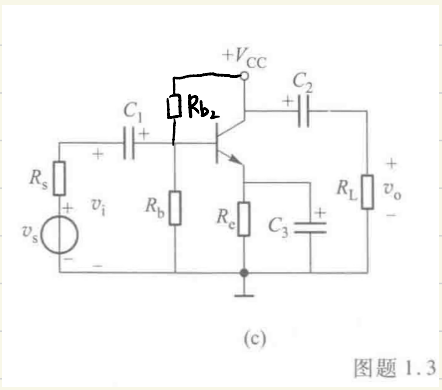
图题 1.3.7

1.3.8

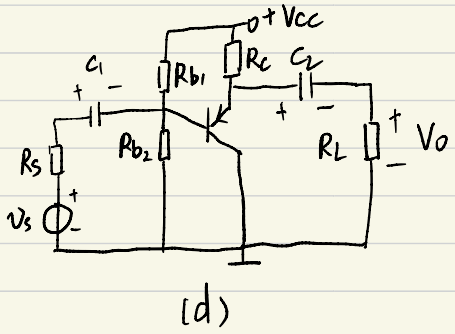




(b)



图题 1.3

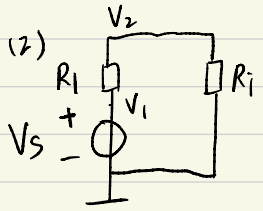


(d)

1.3.10

(1) $20dB+40dB=60dB$

(2)



$$\frac{V_1-V_2}{R_1}=\frac{V_2}{R_i}$$

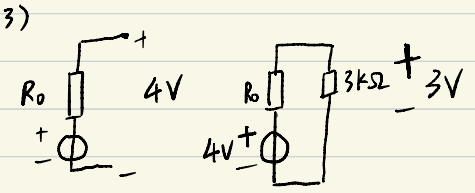
$$R_i=\frac{V_2}{V_1-V_2}R_1$$

万用表内阻使测量偏大

则应选择内阻较大(>1MΩ)的万用表

$$R_i=\frac{V_2}{V_1-V_2}1k\Omega$$

(3)



$$i=\frac{3}{3k\Omega}=1mA$$

$$R_0=\frac{4-iR_L}{i}=1k\Omega$$

(4) 示波器同时观察输入信号和传出信号

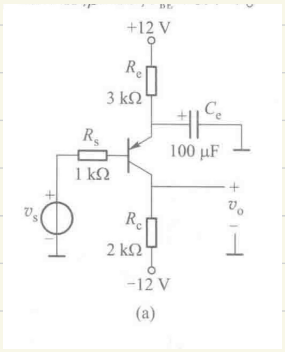
① 信号源输入频率变化时，微调电压，保持测量的输入信号幅值不变

② 毫伏表有良好频率响应，测量幅值

③ 时刻观察示波器，保证不失真

1.3.12

(a)

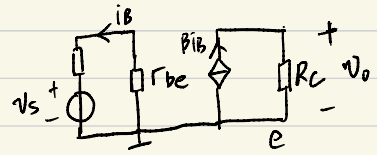


$I_E R_E + I_B R_S + V_{BE} = 12V$

$V_{CE} = -12 - 12 + \beta I_B (R_C + R_E)$

$I_{BQ} = 75\mu A \quad I_{CQ} = 3.75mA$

$V_{CEQ} = -5.25V$



共射组态 (CE)

$r_{be} = r_{bb'} + (1+\beta)\frac{V_i}{I_{EQ}} = 550\Omega$

$A_V = \frac{\beta i_B R_C}{-i_B r_{be}} = -182$

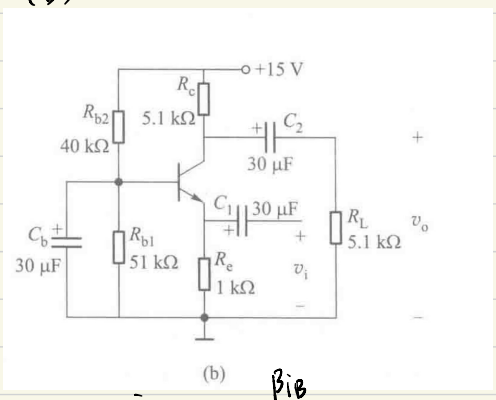
$R_i = \frac{v_i}{i_b} = r_{be} = 550\Omega$

$R_o = \frac{v_o}{\beta i_B} = R_C = 2k\Omega$

$\begin{cases} \Delta V_{CE}^{+} = 2 \times 3.75 = 7.5V \\ \Delta V_{CE}^{-} = 5.25 - 0.7 = 4.55V \end{cases}$

∴ 饱和失真，4.55V

(b)



$\beta I_B R_C + V_{BE} = \frac{R_{b1}}{R_{b1}+R_{b2}} \cdot 15 + 0.7$

$I_{BQ} = 18\mu A$

$I_{CQ} = 0.9mA \quad V_{CEQ} = 9.51V$

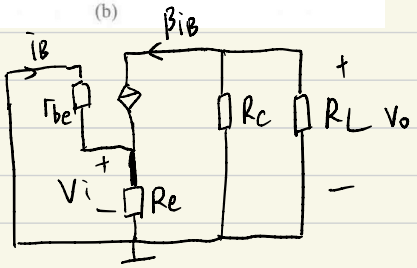
$V_{CE} = 15 - 6.1 I_C$

$R_L' = R_C // R_L$

$\begin{cases} \Delta V_{CE}^{+} = I_{CQ} R_L' = 2.3V \\ \Delta V_{CE}^{-} = 8.81V \end{cases}$

∴ 截止失真

$V_{om} = 2.3V$



CB 组态

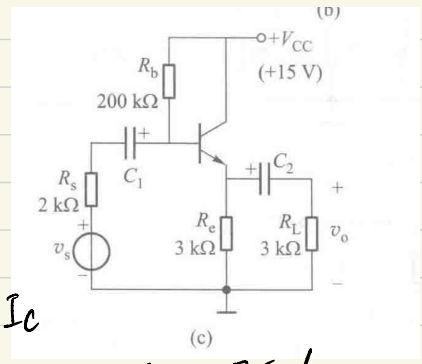
$r_{be} = 1.5k\Omega$

$A_V = \frac{-\beta i_B (R_L // R_C)}{-i_B r_{be}} = 76$

$R_i = \frac{-i_B r_{be}}{(1+\beta) i_B} = 32\Omega$

$R_o = R_L = 5.1k\Omega$

(c)



$15 = I_B R_b + V_{BE} + \beta I_B R_e$

$I_{BQ} = 41\mu A \quad I_{CQ} = \beta I_{BQ} = 2.05mA$

$V_{CEQ} = 8.85V$

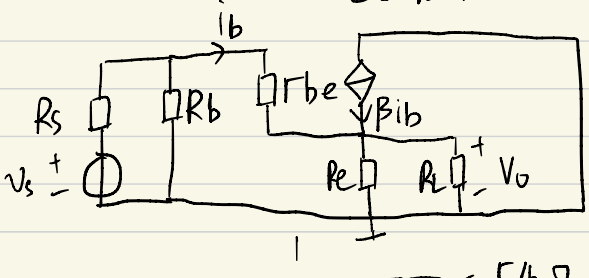
$V_{CE} = 15 - 3 I_C$

$\Delta V_{CE}^{+} = 2.05 \times 1.5 = 3.075V$

$\Delta V_{CE}^{-} = 8.15V$ ∴ 截止失真

$V_{om} = 3.075V$

CC 组态



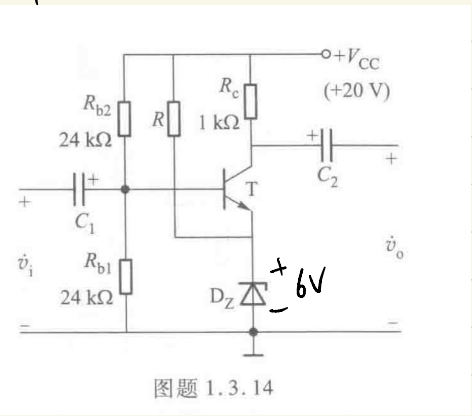
$r_{be} = 834\Omega$

$A_V = \frac{(1+\beta) i_b (R_e // R_L)}{i_b r_{be} + (1+\beta)(R_e // R_L)} = 0.99$

$R_i = R_b // \left[r_{be} + (1+\beta)(R_e // R_L) \right] = 55.8k\Omega$

$R_o = \frac{1}{\frac{1}{R_e} + \frac{1+\beta}{r_{be} + R_S // R_b}} = 54\Omega$

1.3.14



图题 1.3.14

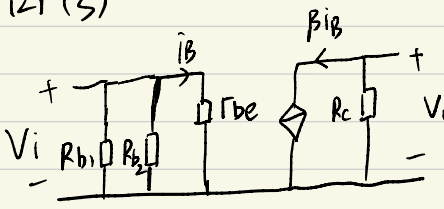
(1) $\frac{20-0.7}{R_{b_2}} - \frac{0.7}{R_{b_1}} = I_B$

$I_{BQ} = 275\mu A$

$I_{CQ} = 5.5mA$

$V_{CEQ} = 20 - 6 - 5.5 = 8.5V$

(2) (3)



$r_{be} = 0.4k\Omega$

$\dot{A}_V = \frac{-\beta i_B R_c}{i_B r_{be}} = -50$

$R_i = R_{b_1} // R_{b_2} // r_{be} = 387\Omega$

$R_o = R_c = 1k\Omega$

(4) 使 Dz 具有更好的稳压性能

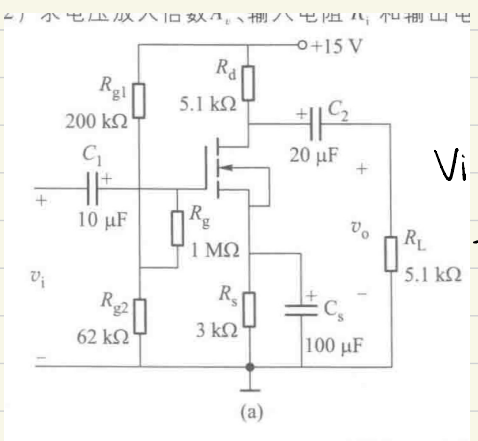
(5) Dz 反接 $\frac{20-0.7}{R_{b_2}} - \frac{0.7}{R_{b_1}} = I_B$

$I_{BQ} = 775\mu A$ $I_{CQ} = 15.5mA$

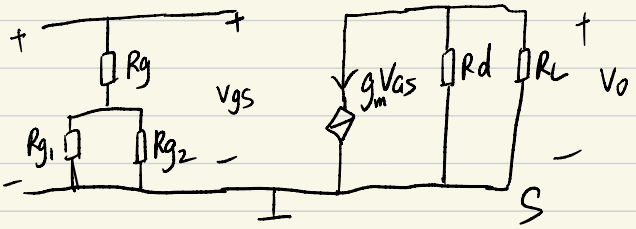
$V_{CEQ} = 4.5V$ 仍在放大区正常工作

$\begin{cases} \dot{A}_V \text{ 增大} \\ R_i \text{ 减小} \end{cases}$

1.3.15 (a)



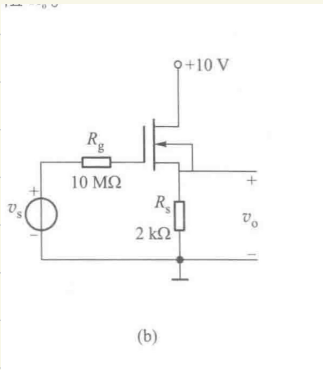
共源 CS 组态



$\dot{A}_V = \frac{-g_m V_{gs}(Rd // R_L)}{V_{gs}} = -5.1$

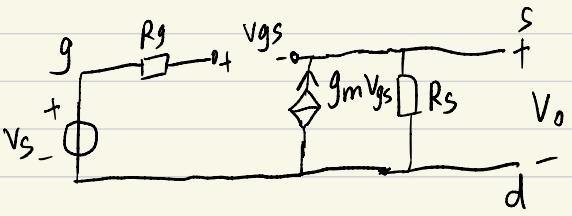
$R_i = 1.047M\Omega$ $R_o = (R_L \to \infty) // Rd = Rd = 5.1k\Omega$

(b)



CD 共漏组态

I: d→s



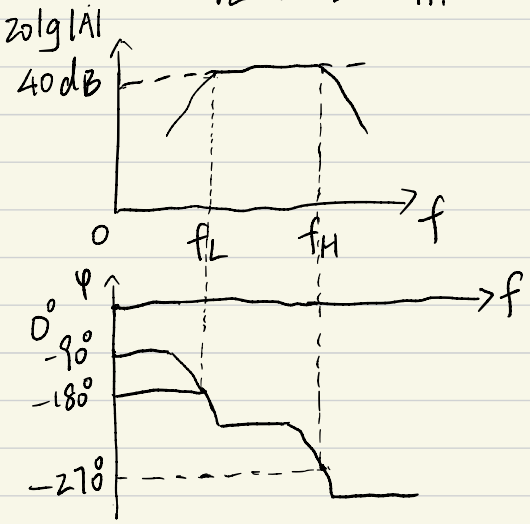
$$\dot{A}v = \frac{g_m V_{gs} R_s}{V_{gs} + g_m V_{gs} R_s} = 0.8$$

$$R_i = \infty \quad (I_i = 0)$$

$$V_o = \frac{V}{\frac{V}{R_s} + g_m V} = 0.4k\Omega$$

1.45 $$\dot{A}v = -100 \cdot \frac{jf/10Hz}{1+jf/10Hz} \cdot \frac{1}{1+jf/50KHz}$$

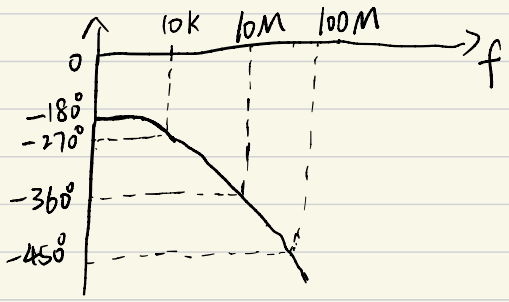
$\therefore f_L = 10Hz \quad f_H = 50KHz$



$\varphi = -180° + 90° - \arctan(f/10Hz) - \arctan(f/50KHz)$

1.47 (1) $$\dot{A}v = -10^4 \cdot \frac{1}{(1+jf/100kHz)(1+jf/10MHz)(1+jf/100MHz)}$$

(2) $\varphi = -180° - \arctan(f/100K) - \arctan(f/10M) - \arctan(f/100M)$



2.1.9 (1) 将 R_{b1}、R_{b2}、V_{CC} 戴维南等效后

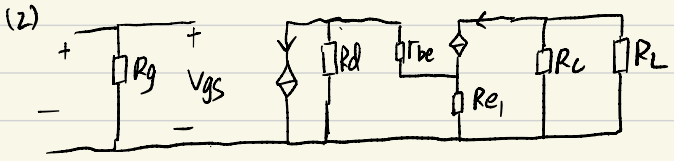
$V_{CC}' = \frac{15}{4} V$　$R_b = 15k\Omega$

$V_{CC}' = I_{BQ}R_b + V_{BE} + (1+\beta) I_{BQ} (R_{e1}+R_{e2})$

$I_{BQ} = 25.4\mu A$　$I_{CQ2} = \beta I_{BQ} = 1.27mA$

$V_{CEQ2} = V_{CC} - I_{CQ2}(R_c + R_{e1} + R_{e2}) = 8.52V$

(2)



(3) $\dot{A}_{V1} = -g_m Rd // Rb // [r_{be} + (1+\beta) R_{e1}]$

$\dot{A}_{V2} = \frac{-\beta(R_c // R_L)}{r_{be} + (1+\beta)R_{e1}}$

$\dot{A}_V = \dot{A}_{V1} \cdot \dot{A}_{V2} = 74.4$

(4) $R_i = 47.1 M\Omega$

$R_o = R_L = 3k\Omega$

(5) $\begin{cases} \Delta V_{CE}^{+} = 1.905V \\ \Delta V_{CE}^{-} = V_{CEQ} - V_{CES} = 7.823V \end{cases}$

∴ 截止失真

$V_{om} = 1.905V$

2.1.13

(1) T_1 组成 共集电路 (CC)，放大电压

T_2 组成恒流源，提供电源

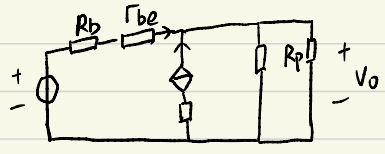
(2) V_{CC}、R_1、R_2 戴维南等效

$V' = \frac{R_2}{R_1+R_2} V_{CC}$　$R' = R_1 // R_2$

$V' = I_{b2} R' + V_{BE} + \beta_2 I_{b2} R_3$

$I_{b2} = 0.16mA$　∴ $I_{C1} = I_{E2Q} = \beta I_{b2} = 8mA$

(3)



$\dot{A}_V = \frac{1}{2} \frac{(1+\beta)R_p}{r_{be} + (1+\beta)R_p} = 0.5$

$R_i = R_b + r_{be} + (1+\beta)R_p = 2613.5k\Omega$

$R_o = \frac{1}{2}R_p // \left(\frac{1}{2}R_p + \frac{R_b + r_{be}}{\beta}\right) = 12.8k\Omega$